AI-Assisted Compiler Error Diagnosis System

Week 13 — System Testing & Validation Report

Phase 7: Testing & Finalization

Student: MADDELA RAJEEV **Roll No:** 24CSB0B40 **Course:** CS1202

Date: March 2026

Project Recap — What Has Been Done (Weeks 1–12)

The following table summarises the deliverables completed across all 12 weeks preceding this testing phase.

Week	Phase	Deliverable	Status
Week 1	Phase 1	Problem Definition, Compiler Workflow Study	Done
Week 2	Phase 1	Literature Survey, Research Gap Identification	Done
Week 3	Phase 2	SRS Document + 8 Error Category Definitions	Done
Week 4	Phase 2	Architecture Diagram, Module Interaction, Tech Stack	Done
Week 5	Phase 3	Error Collection Module — 82 `.cpp` sample files compiled; 129 errors captured in `collected_errors.csv`	Done
Week 6	Phase 3	Feature Extraction — 28 ML features extracted into `features_dataset.csv` (129 rows, 32 columns)	Done
Week 7	Phase 4	Dataset Preparation — Labelled + train/test split; `label_mapping.json` created	Done
Week 8	Phase 4	Model Training — 4 models trained; Random Forest best at 86% accuracy; `best_model.pkl` saved	Done
Week 9	Phase 5	Error Classification Integration — `classify_errors.py` integrates model with compiler output	Done
Week 10	Phase 5	Fix Suggestion Engine — `fix_suggestions.py` + `main.py` providing end-to-end diagnosis output	Done
Week 11	Phase 6	Security & Robustness — 18 tests, 100% pass rate; no dangerous patterns found	Done

Week	Phase	Deliverable	Status
Week 12	Phase 6	Performance Optimisation — Classification accuracy 86% (>85% target); CV mean 87.7%	Done

What Is Left — Remaining Work (Weeks 13–14)

Week 13 — System Testing & Validation (This Week)

According to the project plan, Week 13 requires:

- End-to-end testing using unseen C++ programs
- Comparing standard compiler output vs. AI-assisted output side by side
- Creating formal test cases documenting inputs, expected outputs, and actual outputs
- Producing a validation results report

Week 14 — Documentation & Submission (Next Week)

The final week requires:

- Final project report compilation
- Presentation slides (covering all 14 weeks)
- Source code organisation and cleanup

Week 13 — Testing & Validation (This Deliverable)

This section documents all testing and validation work carried out this week, covering end-to-end system testing with unseen programs and a structured comparison of standard vs. AI-assisted compiler output.

End-to-End Test Cases

The following 16 test cases use programs not included in the training dataset. Each test records the compiler error, AI prediction, confidence, fix suggestion, and whether the output was correct.

Group A: Missing Semicolon

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
TC-01						

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
	`test_ms_new1.cpp`	expected `';'` before `'double'`	Missing Semicolon	97.2%	Add `';'` after statement on error line	PASS
TC-02	`test_ms_new2.cpp`	expected `';'` before `'while'`	Missing Semicolon	95.8%	Add `';'` at end of variable declaration	PASS

Group B: Undeclared Variable

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
TC-03	`test_uv_new1.cpp`	`score` was not declared in this scope	Undeclared Variable	91.3%		

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
					Declare `score` before first use	PASS
TC-04	`test_uv_new2.cpp`	`temp` was not declared in this scope	Undeclared Variable	88.6%	Add `int temp;` before usage	PASS

Group C: Type Mismatch

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
TC-05	`test_tm_new1.cpp`	invalid conversion from `const char*` to `int`	Type Mismatch	96.1%	Assign string to string variable	PASS
TC-06	`test_tm_new2.cpp`					

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
		cannot convert `std::string` to `int`	Type Mismatch	93.4%	Use `std::stoi()` to convert	PASS

Group D: Unmatched Braces

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
TC-07	`test_ub_new1.cpp`	expected `'}'` at end of input	Unmatched Braces	89.5%	Add missing closing brace for if block	PASS
TC-08	`test_ub_new2.cpp`	expected `'}'` before `'return'`	Unmatched Braces	85.1%	Check brace balance; add missing brace	

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
						PASS

Group E: Unmatched Parentheses

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
TC-09	`test_up_new1.cpp`	expected `')'` before `'{'` token	Unmatched Parentheses	90.8%	Add missing `')'` in if condition	PASS
TC-10	`test_up_new2.cpp`	expected `')'` before `';'` token	Unmatched Parentheses	79.2%	Add missing `')'` in for-loop condition	FAIL[*]

[*]Misclassified as Unmatched Braces.

Group F: Missing Return Statement

TC#	**Input File**	**Compiler Error**	**Predicted**	**Conf.**	**Fix Suggested**	**Result**
TC-11	`test_mr_new1.cpp`	control reaches end of non-void function	Missing Return Statement	98.0%	Add return statement at end of function	PASS
TC-12	`test_mr_new2.cpp`	control reaches end of non-void function	Missing Return Statement	97.1%	Add `return 0;` after last statement	PASS

Group G: Invalid Declaration

TC#	**Input File**	**Compiler Error**	**Predicted**	**Conf.**	**Fix Suggested**	**Result**
TC-13	`test_id_new1.cpp`	unknown type name `integr`	Invalid Syntax in Declarations			

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
				94.3%	Fix typo: `integr` → `int`	PASS
TC-14	`test_id_new2.cpp`	two or more data types in declaration	Invalid Syntax in Declarations	82.6%	Remove duplicate type keyword	PASS

Group H: Missing Header Include

TC#	Input File	Compiler Error	Predicted	Conf.	Fix Suggested	Result
TC-15	`test_mh_new1.cpp`	`abs` was not declared in this scope	Missing Header Include	86.9%	Add `#include <cstdlib>` at top	PASS
TC-16						

TC# **Input File** **Compiler Error** **Predicted** **Conf.** **Fix Suggested** **Result**
`test_mh_new2.cpp`
`ceil` was not declared in this scope
Missing Header Include
84.2%
Add `#include <cmath>` at top
PASS

Test Case Validation Summary

Error Category	**Tests**	**Passed**	**Failed**	**Pass Rate**
Missing Semicolon	2	2	0	100%
Undeclared Variable	2	2	0	100%
Type Mismatch	2	2	0	100%
Unmatched Braces	2	2	0	100%
Unmatched Parentheses	2	1	1	50%
Missing Return Statement	2	2	0	100%
Invalid Syntax in Declarations	2	2	0	100%
Missing Header Include	2	2	0	100%
TOTAL	**16**	**15**	**1**	**93.75%**

Note: The single failure (TC-10) was a case where an unmatched parenthesis in a for-loop condition was misclassified as Unmatched Braces. This is a known boundary case where features for the two categories overlap — both trigger the `expected` keyword in the error message and involve bracket-type symbols.

Standard Compiler Output vs. AI-Assisted Output Comparison

Example 1 — Missing Semicolon (TC-01)

Aspect	Standard Output	AI-Assisted Output
Raw Message	`test_ms_new1.cpp:6:16: error`: expected `';'` before `'double'`	Same raw message displayed for reference
Category	(Not provided)	Missing Semicolon
Confidence	(Not provided)	97.2%
Explanation	(Not provided)	A statement must end with `;`. The compiler expects one after the expression on line 6.
Fix Steps	(Not provided)	1. Go to line 6 2. Add `';'` at end 3. Recompile
Code Example	(Not provided)	Wrong: `int x = 10` Correct: `int x = 10;`

Example 2 — Missing Return Statement (TC-11)

Aspect	Standard Output	AI-Assisted Output
Raw Message	`test_mr_new1.cpp:8:1: warning`: control reaches end of non-void function	Same raw message displayed for reference
Category	(Not provided)	Missing Return Statement
Confidence	(Not provided)	98.0%
Explanation	(Not provided)	The function is declared to return a value (non-void) but not all execution paths have a return statement.
Fix Steps	(Not provided)	1. Identify return type 2. Add return statement 3. Check all if/else paths
Code Example	(Not provided)	Wrong: `int f(){}` Correct: `int f(){return 0;}`

Overall Validation Results

Metric	Target (SRS)	Achieved	Status
Classification Accuracy (test set)	$\geq 85\%$	86%	PASS

Metric	Target (SRS)	Achieved	Status
Cross-Validation Mean Accuracy	$\geq 85\%$	87.7%	PASS
Response Time (per file)	< 3 seconds	< 1 second	PASS
End-to-End Test Pass Rate (unseen)	$\geq 85\%$	93.75% (15/16)	PASS
Memory Usage	< 500 MB	< 150 MB	PASS
Robustness Tests (Week 11)	18/18 pass	18/18 (100%)	PASS
Security — Dangerous Patterns	0 patterns	0 patterns	PASS
Error Categories Supported	8 categories	8 categories	PASS

Conclusion

Week 13 end-to-end testing confirms that the AI-Assisted Compiler Error Diagnosis System meets or exceeds all requirements defined in the Week 3 SRS document. The system correctly classified 15 out of 16 unseen test programs, achieving a 93.75% pass rate on completely new inputs — exceeding the 85% accuracy target.

The one misclassification (TC-10) highlights a known boundary between Unmatched Parentheses and Unmatched Braces, where overlapping error message features make the two categories harder to distinguish. This is flagged for documentation in the Week 14 final report as a known limitation with a potential improvement path.

The system is now validated and ready for Week 14 final documentation and submission.